

Higher UK Long-Term Interest Rates

Small-open-economy NK with a HANK-lite mortgage block

Seminar notes on the `uk-long-rate` MVP

Open-source notes on the signed sketch and the calibrated twin
not a Bayesian estimate

`uk-long-rate/` in MacroModellerBot

September 2026

Sources: signed channels review; model sketch v3 (21 Sep 2026);
Dynare twin `dynare/uk_long_rate.mod` and the Python QZ paths in `output/`.

Outline

- 1 The question
- 2 Literature
- 3 The model
- 4 Results
- 5 Next steps
- 6 Appendix

Channels from the signed review first, then the calibrated twin. Every results path below is that twin. Bayesian estimation has not been run.

What we are asking

Question (signed sketch)

How does an exogenous rise in UK long-term nominal rates transmit to GDP, CPI, house prices, sterling, consumption, investment and the public interest bill, with the short policy-rate path held fixed or allowed to move?

Three shocks, kept separate. Illustrative size in the model: a +100 bp move in the model 10-year rate.

- **(a) sterilised term premium.** The term premium lifts the long rate. The expected Bank Rate path is held fixed.
- **(b) path news.** News about future short rates. Bank Rate is not pegged.
- **(c) sovereign.** A fiscal / sovereign gilt risk premium. **Not yet modelled.**

Why “the 10-year yield” is not one shock

- A long yield is an expected path of short rates plus a term premium (duration, inflation, liquidity).
- A path-driven rise often co-moves with stronger expected activity or inflation. A term-premium rise — gilt supply, QE/QT, fiscal credibility, risk-off — tightens more like a cost of capital (Kaminska, Mumtaz and Šustek, 2021; Greenwood, Hanson, Stein and Sunderam, 2023).
- High-frequency surprises are not the same object as the gradual 2021–23 repricing, which was partly anticipated and front-loaded into mortgages (Burr and Willems, 2024).
- For the UK, long rates price fixed-rate mortgages (via swaps), the public interest bill, and an open-economy portfolio balance into sterling.

Do not treat “10-year yield up 100 bp” as interchangeable across **(a) sterilised term premium**, **(b) path news** and **(c) sovereign**.

Path versus term premium

- **Comparable UK macro IRFs** for a pure term-premium shock versus a Bank Rate / path shock are sparse (signed review, §1.1).
- UK *path* peaks are the high-frequency Bank Rate / MPC studies. UK *term-premium* evidence is much better for *yields* (Joyce and Lengyel, SWP 1,097) than for full macro IRFs.
- Kaminska, Mumtaz and Šustek (SWP 914, 2021; also *JME*) is a **US** FOMC decomposition. Useful for the *shape* of path versus premia responses. Instruments are normalised differently. Not a UK calibration target.
- The review's modelling implication, which this MVP follows: separate **(a) sterilised term premium**, **(b) path news** and **(c) sovereign**. Empirically, sterilise (a) by residualising long-end surprises on Bank Rate / short OIS.

Magnitudes, with the shock labelled

Object	Source	Peak / key magnitude
Bank Rate / MPC surprise	Cesa-Bianchi, Thwaites and Vicondoa (2020); ranges in Burr and Willems (2024). UK.	GDP roughly -0.5% to -1.4% ; prices -0.3% to -1.0% ; unemployment $\sim +0.5$ pp in some estimates; houses $\sim -4\%$; sterling ERI $\sim +2\%$ to $+7\%$. Per ≈ 1 pp. Path , not pure term premium.
Housing share of consumption	Albuquerque, Lazarowicz and Lenney, SWP 1,115 (2025). UK.	Housing channel $\approx$ one half of peak transmission to household consumption, among mortgagors. Path / rate shock.
Gilt supply / QT-sized surprise, 10-year yield	Joyce and Lengyel, SWP 1,097 (2024). UK.	~ 20 bp in low stress; ~ 120 bp in high stress. Yield impact, not a macro IRF. Term premium / gilt supply .
Action, expected path, uncertainty	Kaminska, Mumtaz and Šustek, SWP 914 (2021). US.	Action: NK contraction. Path: often expansionary IP. Premia: yields and mortgage rates up, housing down. Shapes only.

Condensed from the signed review, §1.1. Passive QT runoff $\approx$ active sales, pound for pound, in SWP 1,097.

Mortgage cash flow: 2-year and 5-year fixes

- New UK fixes are priced off sterling swaps, which co-move with gilts. A gilt spike can reprice the *flow* between MPC meetings (Burr and Willems, 2024).
- The *stock* is slow. The share of outstanding mortgages fixed for five years rose from around **20% in 2016 to over 50% in 2022** (Burr and Willems, 2024, fn. 18).
- Over 90% of *new* mortgages have been fixed-rate since around 2013, with five-year products replacing two-year fixes (Burr and Willems, 2024; MLAR flow evidence).
- Cash-flow peaks should lag the peak in flow mortgage rates, over a **2–5 year** reset ladder. Anticipation of a dearer refinance can move some spending earlier (Anjum and Herler, 2023, *cited via* Burr and Willems).
- Effective stock rates rose more slowly in 2021–23 than in past cycles because of that fixed share (IMF, 2024).

SWP **1,158** (Rajan, Rodriguez-Tous and Salgado-Moreno, 2025) studies how term *spreads* shift the supply of short versus long fixes. It is not a source for stock reset shares.

Housing collateral, alongside the cash-flow channel

- Higher long rates raise the user cost of housing, cut the present value of housing services, and tighten LTVs when prices fall (Mishkin, 2007; IMF WEO, April 2024, Ch. 2).
- BoE staff: a 1 pp **Bank Rate** hike lowers UK house prices by roughly 4% on average (Burr and Willems, 2024). That is a **path** magnitude, not a term-premium IRF.
- SWP **1,115**: the housing channel is about half of peak transmission to household consumption, concentrated among mortgagors. In their high-frequency evidence, house prices trough around one year; nominal rents are stable for one to two years, then decline.
- Housing wealth effects on consumption are typically at least as large as equity wealth effects (Case, Quigley and Shiller, 2001). UK MPCs out of wealth are cited at around 5p per £1 (Slacalek, 2009, *cited via* Burr and Willems).
- Direct intertemporal substitution looks small next to cash flow, income and wealth (Havranek et al., 2015; Best et al., 2020, for the UK EIS).

Investment, and corporate credit versus gilt premia

- User cost rises with the real financing rate (Hall and Jorgenson, 1967). Long-lived capital is the usual margin; timing can also respond to short rates (McKay and Wieland, *cited via* Burr and Willems, 2024).
- Gilchrist and Zakrajšek (2007): a 1 pp rise in user cost cuts the investment rate by about 50–75 bp in the short run (roughly 30–50 bp from a 1 pp rise in the real long rate), with a long-run capital-stock elasticity near one.
- UK firm evidence in the hiking cycle: higher financing costs plus weaker demand (Burr and Willems, 2024; Shah et al., 2024, *cited via* them). Internal cash buffers dampen the response (Joseph et al., 2019, *cited via* them). Corporate-bond pass-through settled near ~ 0.7 (IMF UK Selected Issues, 2024).
- A gilt term premium is not an excess bond premium. Gilchrist and Zakrajšek (2012) split US spreads into expected default and an EBP; Gertler and Karadi (2015) stress credit costs as well as the funds rate.

The review allows a thin spread block, or an explicit deferral. This MVP defers IG/HY credit (and CRE). Transmission runs through gilt / swap user costs.

The public interest bill

- Higher long yields raise debt service on new and refinanced gilts, and bond supply absorbed by risk-averse investors can crowd out private investment (Greenwood and Vayanos, 2014).
- Joyce and Lengyel (SWP **1,097**, 2024): issuance surprises move the curve through duration-risk and local-supply channels. A fully unexpected QT-sized programme: about **20 bp** at the 10-year in low stress, nearly **120 bp** in high stress. Passive runoff and active sales are broadly similar, pound for pound.
- IMF (2024): gilt yields were more responsive to policy-rate increases than in past UK cycles, consistent with elevated debt and post-mini-Budget risk perceptions.
- With elevated public debt, a term-premium shock has a first-order fiscal leg, and that leg can feed back into confidence and gilt-market functioning.

Modelling implication from the sketch: keep an index-linked share. Do not set $s^{ll} = 0$.

Sterling depends on the shock; LDI is not linear

FX sign (review §2.5)

- A 1 pp **Bank Rate** surprise: sterling ERI about +2% to +7%, peaking within about a quarter (Burr and Willems, 2024; Forbes et al., *cited via* them).
- **(a) sterilised term premium:** a relative-rate reading. Sterling tends to **appreciate**.
- **(c) sovereign** (review only): gilt–FX risk, so the same long-rate rise co-moves with a **depreciation**. Not yet modelled here.

Do not hard-wire “higher long rates $\Rightarrow$ a stronger pound”
(Greenwood et al., 2023; Gourinchas and Rey, 2007).

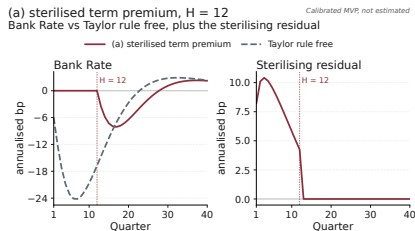
LDI: non-linear stress, off in the MVP

September 2022: leveraged LDI faced NAV and collateral calls after the 23 September mini-Budget; forced gilt sales amplified the sell-off (BoE Financial Policy Summary, 30 Sep 2022; Financial Stability Report, December 2022; Pinter, SWP **1,019**, 2023). Joyce and Lengyel’s ~ 20 versus ~ 120 bp gap is the same stress non-linearity. **Leave it off** for a sterilised, calm-market term premium. Switch it on only under gilt stress, leveraged non-bank duration, or a large sudden credibility shock.

SOE-NK with HANK-lite savers and mortgagors

- Small open economy. Hybrid Phillips curve, Taylor rule, q -theory investment, UIP.
- Two households only: **savers** and **mortgagors**. Not a full idiosyncratic-income distribution.
- Mortgage bridge from the yield curve. One remortgage hazard updates both the stock rate and the LTV debt stock. No second cohort, no buy-to-let.
- Sticky housing user-cost curve, so house prices are not an impact jump. Fiscal interest bill with an index-linked share and a duration refix. Resource constraint with net exports.
- Linear system in `dynare/uk_long_rate.mod`: 19 predetermined variables, 6 forward-looking variables in the declaration, 11 static definitions. Blanchard–Kahn count on this calibration: 36 stable roots, 6 finite unstable (`output/success_checks.txt`).
- The paths in `output/` are the Python QZ twin. `stoch_simul` leaves the Taylor rule free and is **not (a) sterilised term premium**.

Three shocks, and how (a) is sterilised



(a) sterilised term premium: anticipated Bank Rate peg, $H = 12$ quarters (`H_peg` in `src/calibration.py`, `dynare/params_calibration.m`, `dynare/run_uk_long_rate.m`). The Taylor rule stays in the model. For the first H quarters $\varepsilon_t^{\text{ster}} = -i_t^{\text{TR}}$, so

$$i_t = \rho_i i_{t-1} + (1 - \rho_i) \left(\varphi_\pi \pi_t + \frac{1}{4} \varphi_y Y_t \right) + \psi_{\text{nfa}} E_t \text{nfa}_{t+1} + \nu_t + \varepsilon_t^{\text{ster}} = 0.$$

$\varphi_y = 0.125$ is the annual convention, hence the factor $1/4$. Residuals are anticipated; the rule resumes after quarter 12.
(b) path news is not sterilised. **(c) sovereign** is not yet modelled.

New-loan rate versus the mortgage stock

Debt service uses the stock rate. User cost and new borrowing use the new-loan rate.

$$R_t^m = \omega_S i_t + \omega_L R_t^L + \zeta_t, \quad \omega_S + \omega_L = 1,$$
$$R_t^{m,\text{stock}} = (1 - \lambda_q) R_{t-1}^{m,\text{stock}} + \lambda_q R_t^m.$$

- $\lambda_q = 0.09$ is held fixed (sketch). Mean spell $1/0.09 \approx 11.1$ quarters; annual reset probability $1 - 0.91^4 \approx 31\%$. Provisional weights: $\omega_S = 0.30$, $\omega_L = 0.70$.
- Borrower consumption:

$$C_t^b = \alpha_y Y_t - \mu_{ds} R_t^{m,\text{stock}} + \mu_{\text{coll}} \lambda_q (P_t^h - B_{t-1}^m).$$

$\mu_{ds} = 15$, $\alpha_y = 0.15$, $\mu_{\text{coll}} = 0.10$. The collateral term is provisional, and it only bites on the fraction λ_q that remortgages.

Interest bill: index-linked share and duration

Do not double-count s^{ll} . $B^{\text{nom}} = (1 - s^{ll})B^g$ and $B^{ll} = s^{ll}B^g$, with B^g equal to 4 quarters of GDP (about 100% of annual GDP). In the linear units, $b^{\text{nom}} = 3$ and $b^{ll} = 1$.

$$R_t^{\text{eff}} = (1 - \delta_D) R_{t-1}^{\text{eff}} + \delta_D R_t^L, \quad \delta_D = \frac{1}{4D},$$

$$IB_t = b^{\text{nom}} R_t^{\text{eff}} + b^{ll} (r_t^{ll} + \pi_t) + \frac{i_{ss}}{100} \tilde{B}_t^g.$$

- $s^{ll} = 0.25$ and $D = 9.1$ years are held fixed, so $\delta_D = 1/36.4 \approx 0.02747$. On impact, from a zero lag, $R_0^{\text{eff}}/R_0^L = 1/(4D)$.
- The index-linked real coupon refixes toward $R^L - E\pi$. The inflation uplift sits *inside* reported IB . A disinflation therefore cuts the bill on impact even while the conventional coupon is starting to refix.

UIP: a rise in rer is a depreciation

$$rer_t = E_t rer_{t+1} - (i_t - E_t \pi_{t+1}) + \chi_{nfa} nfa_t + \rho_t^{fx}.$$

- $\chi_{nfa} = 0.01$. Net exports: $NX_t = \eta_{rer} rer_t - \eta_y Y_t + \eta_{y^*} Y_t^*$, with $\eta_{rer} = 0.06$ and $\eta_y = 0.18$.
- Under **(a) sterilised term premium**, $i_t = 0$ on the peg while inflation falls, so the real short rate rises and sterling appreciates (rer falls). That is the review's sign for (a). On this path rer is -1.875% on impact (`irf_tp_sterilised.csv`).
- A passive tax rule $\tau_t = \varphi_{dg} \tilde{B}_{t-1}^g$ ($\varphi_{dg} = 0.06$) closes the fiscal loop. $\varphi_\pi = 1.5$, $\beta = 0.995$.

Calibration: three cells held fixed, the rest provisional

	Value	Status	Role
λ_q	0.09	fixed	quarterly remortgage hazard
s^{IL}	0.25	fixed	index-linked share of the gilt stock
D	9.1y	fixed	modified duration
κ	0.004	provisional	NKPC slope. Flat, so a finite peg is not a Fisher spiral
φ_h, κ_c	0.75, 0.055	provisional	housing user cost; loading on C^b
$\kappa_H, \kappa_{\text{level}}$	0.03, 0.05	provisional	housing stock and level anchor
μ_{coll}	0.10	provisional	LTV / collateral, scaled by λ_q
ω_S, ω_L	0.30, 0.70	provisional	Bank Rate and long-rate weights in R^m
$\beta, \varphi_\pi, \rho_i$	0.995, 1.5, 0.86	provisional	discount factor and the Taylor rule
$\varphi_y, \psi_{\text{nfa}}$	0.125, 0.08	provisional	annual output weight; NFA term inside i^{TR}
ρ_{TP}, ρ_ν	0.90, 0.75	provisional	persistence of (a) and of (b)
h_s, σ, ψ^{TP}	0.40, 6, 2	provisional	saver level rule; eis = $(1 - h_s)/\sigma = 0.10$

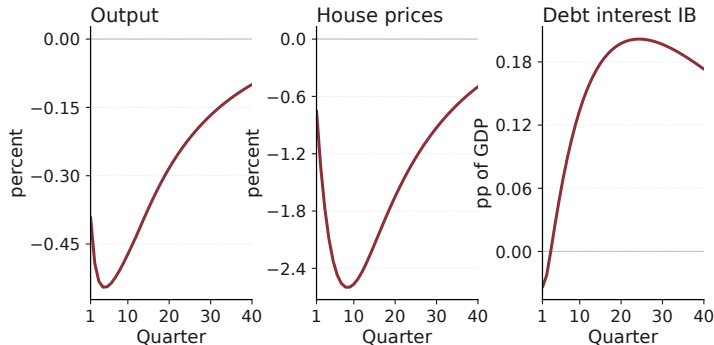
Full list in `src/calibration.py` and the `.mod`. λ_q , s^{IL} and D are not free. κ and the housing loadings are provisional: an earlier pass troughed near -9.5% per $+100$ bp. Not UK estimates.

(a) sterilised term premium

(a) sterilised term premium

+100 bp, anticipated Bank Rate peg $H = 12$

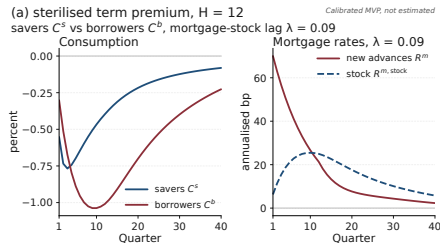
Calibrated MVP, not estimated



Output trough -0.55% in quarter 4. House prices trough -2.6% in quarter 8. Interest bill peaks at $+0.20$ pp of GDP in quarter 24.

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

(a) sterilised term premium: who cuts consumption

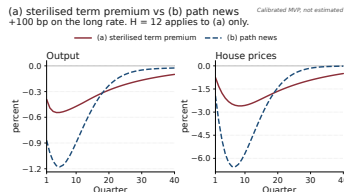


- Savers' consumption falls first: trough -0.77% in quarter 3. It does not rise later (C^s stays in $[-0.77\%, -0.08\%]$).
- Borrowers' consumption falls more, and later: trough -1.04% in quarter 9.
- Only about 9% of the mortgage stock reprices each quarter ($\lambda_q = 0.09$). New-loan R^m jumps 70 bp on impact; the stock rate only 6.3 bp, and peaks in quarter 10, next to the borrower trough.

Exact troughs in `success_checks.txt`: C^s -0.768% at q3, C^b -1.038% at q9.

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

(a) sterilised term premium against (b) path news

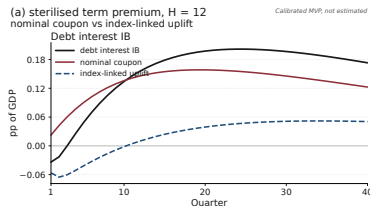


	(a) sterilised term premium	(b) path news
R^L on impact	+100 bp	+100 bp
Bank Rate on impact	pegged at 0 ($H = 12$)	+315 bp
Output trough	-0.55% (q4)	-1.178% (q5)
House-price trough	-2.6% (q8)	-6.551% (q7)

(b), from `irf_short_rate_news.csv`: weight $1/(4D)$ on the current short rate, so Bank Rate peaks at +560 bp in quarter 4. R^L later dips to -11.8 bp in quarter 16 under (b) and stays positive under (a) (minimum +2.35 bp).

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

(a) sterilised term premium: interest-bill split



Columns `ib_nom` and `ib_il` on `irf_tp_sterilised.csv`. *IB* is percentage points of GDP.

- Index-linked piece, including the uplift: -0.0563 pp of GDP on impact (disinflation).
- Nominal coupon: rises immediately, but only at rate $1/(4D)$. Peak $+0.1585$ pp in quarter 19. R^{eff} itself peaks in quarter 17.
- Sum: impact -0.034 pp, peak $+0.20$ pp of GDP in quarter 24 (the check prints $+0.2017$). Coupon half-life is about 25 quarters; $\rho_{TP} = 0.90$ pulls the peak forward of that half-life.

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

Next: high-frequency gilt local projections

`src/lp_proxyvar.py` is a scaffold. It does not ship a surprise file. With no `data/gilt_surprises.csv` it writes `output/lp_irf_stub_Y_Ph.csv` full of NaNs.

- **(a) sterilised term premium**, empirical analogue: the high-frequency 10-year gilt change, residualised on a constant, the Bank Rate surprise, and the near-term OIS path (`orthogonalise_gilt_surprise`).
- **(b) path news**: the OIS path itself.
- Outcomes the stub is shaped for: ONS monthly GDP and a UK house-price index. Horizons 0–12 in the stub. Reference already in the code: Cesa-Bianchi, Thwaites and Viccondoa (2020).
- **(c) sovereign is not yet modelled.**
- **Next estimation step**: a linear Jordà baseline on that residualised surprise, then a state-dependent / non-linear variant à la David, Giacomini, Jiao and Wang. The non-linear variant is not in the repo yet.

No UK surprise series is invented here. A synthetic DGP exists only inside `src/test_lp_algebra.py`.

Then: Dynare Bayesian estimation on UK data

- Not run. The bottom of `uk_long_rate.mod` leaves `estimated_params` for later. λ_q , s^{ll} and D stay fixed.
- A later block may free κ , ρ_{TP} , ρ_ν , ω_S , μ_{ds} and the housing loadings, and only once the observables and the high-frequency instruments are actually loaded.
- **Prior for κ .** Centre it on an import-penetration-dependent Phillips-curve slope à la Hottman and Reyes-Heroles, converted to an aggregate slope, and keep the prior **wide**. The calibrated $\kappa = 0.004$ is a flat slope that stops a finite peg becoming a Fisher spiral. It is not that prior, and it is not a UK posterior.
- Sketch observable list, not yet loaded: Bank Rate / SONIA, the 10-year gilt and gilt-OIS, MLAR 2-year and 5-year fixes, the Bankstats stock rate, UK house prices, ONS consumption, investment, GDP and CPI, the sterling ERI, and the DMO interest bill.

Takeaways

- Separate **(a) sterilised term premium**, **(b) path news** and **(c) sovereign**. UK path-shock macro peaks are not UK term-premium IRFs. LDI / September 2022 stays a boxed non-linear stress effect.
- In the **calibrated** twin, a +100 bp **(a) sterilised term premium** ($H = 12$) lowers output (trough -0.55% in quarter 4) and house prices (trough -2.6% in quarter 8). The interest bill peaks at +0.20 pp of GDP in quarter 24, after an index-linked dip. Savers' consumption falls first (-0.77% in quarter 3) and does not rise. Borrowers fall more, and later (-1.04% in quarter 9), because only about 9% of the mortgage stock reprices each quarter ($\lambda_q = 0.09$).
- **(b) path news**, scaled to the same long-rate impact, is a much larger short-rate path and a deeper contraction. **(c) sovereign** is not yet modelled.
- These are not estimates. κ and the housing parameters are provisional. Next: residualised gilt local projections, including a state-dependent variant, then Bayesian estimation with a wide prior on κ .

Appendix: yield curve, saver rule, housing curve

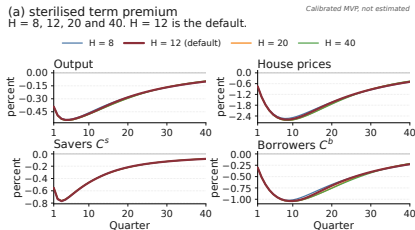
$$R_t^{EH} = \delta_D i_t + (1 - \delta_D) E_t R_{t+1}^{EH}, \quad R_t^L = R_t^{EH} + TP_t.$$

$$C_t^s = h_s C_{t-1}^s - \psi^{TP} TP_t - \text{eis}(i_t - E_t \pi_{t+1}) + \psi^y Y_t + \psi^{IB} IB_t - \psi^\tau \tau_t.$$

$$\begin{aligned} (1 + \beta + \kappa_{\text{level}}) P_t^h &= P_{t-1}^h + \beta E_t P_{t+1}^h + \kappa_c C_t^b \\ &\quad - \varphi_h (R_t^m - E_t \pi_{t+1}) - \kappa_H H_t. \end{aligned}$$

The saver equation is a level rule. A forward habit Euler plus a ΔTP capital-gain term had produced a positive tail in C^s (peak about +0.5% on an earlier pass). $\psi^{TP} = 2 > 0$ books the term premium as a duration loss. Housing loadings are provisional: an earlier pass troughed near -9.5% per +100 bp, which was far too large next to UK Bank Rate responses of a few percent. Equations are the .mod / NOTES.md system.

Appendix: peg length is not doing the work



H	Y trough	P^h trough	max C^s	C^b trough	IB peak	min R^L
8	−0.540% q4	−2.540% q8	−0.082%	−1.024%	+0.2016 q25	+2.38 bp
12	−0.545% q4	−2.597% q8	−0.081%	−1.038%	+0.2017 q24	+2.35 bp
20	−0.549% q4	−2.637% q9	−0.080%	−1.046%	+0.2036 q24	+2.25 bp
40	−0.550% q4	−2.645% q9	−0.079%	−1.048%	+0.2046 q24	+2.11 bp

Only $H = 12$ is **(a) sterilised term premium**. The figure labels the other pegs as robustness. Rounded from output/irf_peg_robustness.csv. Results slides round the $H = 12$ row to output −0.55%, house prices −2.6%, and an interest bill of +0.20 pp. $\kappa = 0.004$ and ψ_{nfa} are provisional.

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

Appendix: the unsterilised term premium is not (a)

Same TP innovation as in the sterilised run, Taylor rule left on. This is what `stoch_simul(e_tp)` does. File: `output/irf_tp_unsterilised.csv`. **Not (a) sterilised**

term premium.	R^L on impact	+94.9 bp (not rescaled to 100)	The endogenous
	Bank Rate	−6.0 bp on impact; trough −24.2 bp in quarter 7	
	Output	impact −0.349%; trough −0.487% in quarter 4	
	House prices	trough −2.212% in quarter 9	
	Saver consumption	stays negative (peak −0.080%)	

easing is small next to **(b) path news**, and the sign pattern for output, house prices and C^s matches the sterilised path. It is still a different experiment: Bank Rate is not pegged.

Calibrated model. Bayesian estimation has not been run. κ and the housing parameters are provisional.

Appendix: what the MVP leaves out

In the linear baseline

- **(a) sterilised term premium** with $H = 12$, **(b) path news**, and the interest-bill block with $s^{ll} = 0.25$.
- One mortgage hazard. Savers on a level rule. Flat provisional Phillips curve. Housing user-cost curve with provisional loadings.

Not yet modelled

- **(c) sovereign.**

Not in the MVP

- LDI / gilt-stress amplification. CRE, buy-to-let, a second mortgage cohort, and a corporate EBP block.
- Bayesian estimation, UK surprise data, and the state-dependent local projection.

Selected references

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Compile: pdf_{latex} uk_long_rate_seminar.tex from uk-long-rate/beamer/.